

# Indian Equity Multi-Agent Prediction System

Phase and checkpoint plan | Draft 1, 7 October 2026 | Figures marked as assumptions are verified in Phase 0

## Fixed design choices

| Item           | Choice                                                                                                                           |
|----------------|----------------------------------------------------------------------------------------------------------------------------------|
| Goal           | Predict Up / Sideways / Down for each stock over the next 10 trading days, with a stop and a target on every call.               |
| Timeframe      | Daily bars, 5 to 10 day holding period. Weekly trend is used only as a filter.                                                   |
| Universe       | NIFTY 100 or 200 with historical constituents. Long or cash only.                                                                |
| Reward to risk | 2:1. Target 2x ATR, stop 1x ATR, time limit 10 days. Risk 1% of capital per trade. 1.5:1 and 3:1 tested on validation data only. |
| Labels         | Whichever limit is hit first: target = Up, stop = Down, time-out = Sideways.                                                     |
| Costs          | About 0.3% per round trip in every backtest (assumption).                                                                        |
| Losses         | Zero loss is not achievable. The system delivers a fixed, small loss per trade.                                                  |

## Working rules

- **Gate rule.** No checkpoint moves forward until its check passes. A failed part is fixed or dropped.
- **One report per checkpoint.** Each checkpoint saves a short report, so an error points to one place.
- **Ask first.** Every decision listed under Approval points is put to you before it is acted on.
- **No vote adding.** Agents output calibrated probabilities; a stacking model combines them, so two correlated weak signals do not become one strong signal.

## Indicator families

Each family is tested alone, then as an add-on to the others. A family stays only if it improves out-of-sample results after costs in most walk-forward folds. Several families can pass.

| Family            | Examples                                           |
|-------------------|----------------------------------------------------|
| Trend             | Moving-average slopes, ADX, weekly trend           |
| Momentum          | RSI, rate of change, MACD                          |
| Volatility        | ATR, Bollinger width                               |
| Volume            | OBV, NSE delivery percentage                       |
| Relative strength | Stock vs sector, sector vs Nifty (four quadrants)  |
| Price structure   | Distance from 52-week high, support and resistance |
| Market regime     | Nifty trend, India VIX, breadth, FII/DII flows     |
| Derivatives       | Open interest, put-call ratio, expiry week         |

**RSI.** Included in the momentum family and tested like every other indicator. It is read with trend context: RSI above 70 in an uptrend is strength, not a sell signal. Relative strength versus an index is a separate measure and sits in its own family.

**Combined setups.** An indicator that is weak alone is not thrown away immediately. Indicators are also tested together as one setup, inside a family or across families (for example RSI pullback + rising trend + above-average delivery volume). A setup is kept only if it beats each of its parts alone, appears often enough to trust (minimum sample count fixed in Phase 0), and holds across walk-forward folds. A passed setup enters the model as a single

feature, so it is counted once.

## Phases and checkpoints

### Phase 0: Spec

| No. | Checkpoint                                                             | Pass check                |
|-----|------------------------------------------------------------------------|---------------------------|
| 0.1 | Base paper and baselines chosen                                        | Shortlist agreed with you |
| 0.2 | NSE costs and trading rules verified                                   | Each figure has a source  |
| 0.3 | Universe, horizon, metrics, holdout dates, setup sample minimum frozen | Written spec signed off   |

### Phase 1: Data

| No. | Checkpoint                                           | Pass check                   |
|-----|------------------------------------------------------|------------------------------|
| 1.1 | Prices for the universe with historical constituents | No gaps or missing stocks    |
| 1.2 | Split and bonus adjustment                           | No unexplained one-day jumps |
| 1.3 | Sector mapping and sector index data                 | Every stock has one sector   |
| 1.4 | Quality report and frozen data snapshot              | Snapshot versioned           |

### Phase 2: Labels and backtest engine (before any model)

| No. | Checkpoint                                                   | Pass check                                                            |
|-----|--------------------------------------------------------------|-----------------------------------------------------------------------|
| 2.1 | Target / stop / time labels                                  | Class balance; 20 labels verified by hand on charts                   |
| 2.2 | Backtest engine with costs and next-day execution            | Buy-and-hold matches the index; random signals lose roughly the costs |
| 2.3 | Walk-forward splits                                          | No label window overlaps train and test                               |
| 2.4 | Baselines: buy-and-hold, momentum, plain RSI rule, MACD rule | Numbers to beat recorded                                              |

### Phase 3: Indicators

| No. | Checkpoint                                             | Pass check                                                       |
|-----|--------------------------------------------------------|------------------------------------------------------------------|
| 3.1 | Indicator library                                      | Values match TA-Lib                                              |
| 3.2 | Leak test                                              | Features on truncated data equal features on full data           |
| 3.3 | Family gate, each family alone                         | Out-of-sample gain after costs in most folds                     |
| 3.4 | Combined setups, within and across families            | Setup beats each part alone, enough samples, stable across folds |
| 3.5 | Correlation clustering; one representative per cluster | No pair above the agreed correlation limit                       |
| 3.6 | Add-on test, then freeze the feature set               | Each kept family or setup adds over the rest                     |

## Phase 4: Technical model

| No. | Checkpoint                       | Pass check                                       |
|-----|----------------------------------|--------------------------------------------------|
| 4.1 | Logistic regression as the floor | Beats random                                     |
| 4.2 | Gradient-boosted trees           | Beats 4.1                                        |
| 4.3 | Probability calibration          | Predicted probability matches observed frequency |
| 4.4 | Shuffled-label test              | Accuracy drops to chance                         |
| 4.5 | Backtest after costs             | Compared with 2.4 baselines                      |
| 4.6 | Sequence model                   | Kept only if ablation shows a gain               |

## Phase 5: Sector and regime

| No. | Checkpoint                                             | Pass check                       |
|-----|--------------------------------------------------------|----------------------------------|
| 5.1 | Four quadrants: Leading, Weakening, Lagging, Improving | Match known sector runs          |
| 5.2 | Forward returns by quadrant                            | Quadrants separate returns       |
| 5.3 | Market regime filter                                   | Cuts drawdown in falling markets |
| 5.4 | Added to the model                                     | Ablation shows a gain            |

## Phase 6: News agent

| No. | Checkpoint                                                                        | Pass check                                   |
|-----|-----------------------------------------------------------------------------------|----------------------------------------------|
| 6.1 | Collection with correct timestamps                                                | After 15:30 counts for next day              |
| 6.2 | News-to-ticker mapping                                                            | Precision on 200 hand-labelled items         |
| 6.3 | Event classifier (CEO exit, order win, results, regulatory action, rating change) | Accuracy on labelled set vs FinBERT baseline |
| 6.4 | Event study per event type                                                        | Average move and sample count per type       |
| 6.5 | News agent alone on event days                                                    | Backtest after costs                         |

## Phase 7: Risk agent

| No. | Checkpoint                         | Pass check                            |
|-----|------------------------------------|---------------------------------------|
| 7.1 | Position sizing                    | Loss per trade stays at 1% of capital |
| 7.2 | Veto rules, each tested separately | Cuts drawdown without killing return  |
| 7.3 | Portfolio and sector limits        | Limits never breached in backtest     |

## Phase 8: Fusion

| No. | Checkpoint                                    | Pass check                                                                             |
|-----|-----------------------------------------------|----------------------------------------------------------------------------------------|
| 8.1 | Out-of-sample predictions from every agent    | No in-sample predictions used                                                          |
| 8.2 | Stacking model                                | Beats best single agent and a plain average                                            |
| 8.3 | Double-counting test with a duplicated signal | Output does not strengthen                                                             |
| 8.4 | Past-pattern column                           | Shows count, average 10-day return, hit rate, or 'no related past data' below 30 cases |

## Phase 9: Final validation

| No. | Checkpoint                                                      | Pass check                          |
|-----|-----------------------------------------------------------------|-------------------------------------|
| 9.1 | Holdout                                                         | Run once only                       |
| 9.2 | Ablations and significance tests                                | Every agent's contribution measured |
| 9.3 | Stress periods: March 2020, 2022, June 2024 election result day | Drawdown within limit               |
| 9.4 | Paper trading, 2 to 3 months                                    | Live results in line with backtest  |

## Phase 10: Dashboard and paper

| No.  | Checkpoint                                                                      | Pass check                              |
|------|---------------------------------------------------------------------------------|-----------------------------------------|
| 10.1 | Dashboard: call, probability, stop, target, sector quadrant, news, past pattern | Numbers match the backtest engine       |
| 10.2 | Paper draft against the base paper                                              | Every figure reproducible from the repo |

## Approval points (you decide)

- Base paper and target journal (0.1).
- Universe size and holdout dates (0.3).
- Which families and combined setups are kept or dropped after the gate results (3.3, 3.4, 3.6).
- Whether a sequence model is added (4.6).
- News sources and event types (6.1, 6.3).
- Veto rules and risk limits (7.2, 7.3).
- Any change to reward-to-risk, horizon or costs after Phase 0.

This is a research design, not financial advice. Predictions will be wrong some of the time; the plan controls the size of losses, it does not remove them.